Naming Vertebrae Without an Anchor: Registration-Free Level Identification on Limited-Field Radiographs

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Article type: Original Research

Manuscript word count (Introduction–Discussion): ≈2,600

Abstract word count: 249

Figures: 1 **Tables:** 3 **Funding:** none declared.

Conflicts of interest: the author has a commercial interest in a level-confirmation product under development.

Data sharing: every public source is named in Table 1 with its licence class; derived label tables and every result file are released with the code, so that each reported number can be regenerated. VinDr-SpineXR and AASCE require the reader's own data-use agreement, and results that depend on them are marked. Trained model weights are not released.

Code: the analysis code, including the single-command reproduction chain (`slurm/reproduce_all.sh`), is held in a private repository during review. Access is provided to editors and reviewers on request, and the repository is made public on publication.

Abstract

Purpose: To determine whether a short series of overlapping fluoroscopic exposures can supply the information a single limited field lacks for naming a vertebra, without registration to a preoperative CT.

Materials and Methods: Retrospective study of public data. A ResNet-18 encoder over fixed-scale vertebra crops with a bidirectional recurrent head over 2–9 vertebrae was trained on eleven sources and evaluated on spines excluded from training, with the exclusion enforced by the evaluation tool. Frames spanning a 9-inch intensifier field (14 cm at the spine) were named twice: on the frame's own posterior, and after linking overlapping frames into one chain and decoding under the constraint that the chain is consecutive and terminates at a landmark. Accuracy is the share of vertebrae given their exact name, with Wilson intervals and film-clustered bootstrap.

Results: On 120 held-out abdominopelvic renders, a two-vertebra frame containing no landmark was named exactly 72.1% of the time alone and 95.2% on the chain; on 25 held-out real lumbar radiographs, 78.7% and 93.6%. The effect was largest where a single frame carries least: swept from C2 on 12 held-out whole-spine studies, a two-vertebra thoracic frame rose from 9.5% to 100%. Chains completed on 97.5% of spines at two bodies of overlap and 3.4 exposures, and the protocol as a radiographer would perform it—a fixed move of one field minus two bodies from the sacrum, with no final move shorter than half a body—completed every chain (120 of 120). On 522 clinical CTs from another institution that no model here was trained on, the same protocol completed 98.5% of chains and named a two-vertebra frame 94.3% of the time from 51.2% alone; swept from C2 on 203 more, 95.1% and 94.0% from 18.9%. A sixth lumbar vertebra was named correctly 57–80% of the time and the chain did not rescue it.

Conclusion: The limitation is in the exposure rather than in the classifier. A sweep of overlapping frames is an intraoperative registration built from radiographs instead of from a preoperative CT, and it recovers most of what one frame cannot hold—except on transitional spines, where the corpus, not the method, is the constraint.

Summary Statement: Linking a short series of overlapping fluoroscopic exposures into one chain raised exact vertebral naming of a two-vertebra field from 70.7% to 93.4% on renders and from 13.4% to 97.1% through the thoracic spine, with no preoperative CT.

Key Points

- A limited fluoroscopic field with no landmark in view was named exactly 72.1% of the time from its own pixels and 95.2% when the same posteriors were decoded under the ordering constraint supplied by a chain of overlapping frames (120 held-out spines).
- The gain was largest in the thoracic spine, where adjacent vertebrae most resemble one another and counting from a landmark is standard practice: 9.5% to 100% at two vertebrae in view, on 12 held-out spines.
- Two bodies of overlap and about three exposures completed the chain on 96.7% of spines; beyond that, additional frames reduced accuracy because each is another link that can be joined wrongly.

Abbreviations: CI = confidence interval; CT = computed tomography; DRR = digitally reconstructed radiograph; FOV = field of view; LSTV = lumbosacral transitional vertebra; TLTV = thoracolumbar transitional vertebra.

Introduction

Wrong-level spine surgery persists because intraoperative fluoroscopy shows a few vertebrae at a time and the surgeon counts from whatever landmark is visible [1, 2]. Half of responding neurosurgeons in two independent surveys report at least one wrong-level operation in their career [3, 4]; self-reported incidence is one in 3,110 procedures [3], and a prospectively collected registry of 34,544 degenerative spine operations recorded 0.12 %, with the thoracic spine highest at 0.42 % (95 % CI 0.16–1.08), three times the lumbar and eight times the cervical rate [5]. Published wrong-site rates across all of surgery span 0.09 to 4.5 per 10,000 [6], and a systematic review found twelve incidence studies for the spine, none of them thoracic [7]. The consequences are not confined to the patient. Seventeen percent of self-reported events led to legal action or settlement [3]; nearly one in five Spine Section respondents has been sued for a wrong-level or wrong-side lumbar operation [4]; of 68 litigated wrong-disc-level cases, 55 ended in a payment, and surgeons prevailed in 42 % of those a jury decided [8]. The median indemnity for a surgical never event reported to the National Practitioner Data Bank is $75,220 (interquartile range $28,847–$215,003, 2025 dollars) [9], and in England 978 spinal-surgery negligence claims over five years were valued at £535.5 million, 26 of them never events including wrong-site surgery [10].

The thoracic spine also reveals the difficulty in what is spent avoiding it. One quaternary centre performed 179 preoperative percutaneous gold-fiducial placements over 38 months, 176 of them thoracic, at a mean of 51.9 minutes and 183 mGy each, purely to mark a level for the surgeon [11]. At a hospital operating-room cost of $34–37 per minute before physician fees [12], each placement represents roughly $1,750–$1,950 of staffed procedural time and the series about 155 hours—a proxy, since the marker is placed in a fluoroscopy or CT suite rather than a theatre—and it is a second invasive procedure scheduled in advance because the level was judged un-nameable in theatre.

In theatre the standard technique is unchanged: needles are placed every few levels and counted cephalad from the sacrum on lateral fluoroscopy, or ribs are counted on an anteroposterior film, and surgeons disagree even on which structure to count, the pedicle for 67 % and the spinous process for 49 % of respondents [1, 13]. It is error-prone for reasons the images themselves supply. The field holds three or four bodies, so a count from the sacrum to a mid-thoracic level passes through several exposures each of which can be joined to the last one body off; a lumbosacral or thoracolumbar transitional vertebra shifts the count by one for the whole column; ribs are hard to number where a twelfth rib is absent or hypoplastic; and habitus, osteopenia and

the scapular shadow degrade the mid-thoracic image most [2, 13]. The only study to assess site-verification protocols found they failed to prevent about one-third of cases [7], and the recommended safeguard, imaging after exposure against a marked fixed structure [6], is itself another count.

What that costs in aggregate is bounded from below rather than known. The Joint Commission received 127 reports of wrong surgery in 2024, nine of them wrong-level and seven of those spinal, from a voluntary system it states captures a small fraction of events [14]; claims-based modelling puts surgical never events near 4,000 a year in the United States, a quarter of them wrong-site [15]; NHS England recorded 185 wrong-site never events in a single year [16]. Malpractice payments for all surgical never events summed to $1.3 billion over 21 years [15], and in a contemporary closed-claims series the mean wrong-site indemnity was $136,453 with spine and disc surgery the commonest procedure [17]; applied to the two per-procedure rates above and a volume of 1.3 to 1.6 million spine operations a year, wrong-level spine surgery alone would produce roughly 400 to 1,900 events and $8 to $37 million in indemnity a year, before any hospital or disability cost, none of which has a published per-event figure. Two facts about that burden matter for what follows. It is not removable by checklists: chart review of insured wrong-site cases found the Universal Protocol would have prevented 62 % of them under ideal conditions [18], and wrong-site events continued at a steady rate in the protocol era [19]. And in the spine it is concentrated where counting is longest and the image is worst, the thoracic levels for which a centre pays 52 minutes and 183 mGy per patient to avoid counting at all.

Automated level identification on such images has been approached almost entirely by registering a preoperative CT to the radiograph [20, 21]. That approach imports the vertebra's identity rather than reading it: registration solves a six-degree-of-freedom pose, and the name is transferred from a human's reading of the preoperative volume. Two consequences follow. It is unavailable in the many cases that never had a CT, and a miscounted preoperative scan propagates onto the radiograph faithfully and confidently—which is the failure mode of a spine with transitional anatomy, the population in which wrong-level surgery actually occurs.

Reading identity from the radiograph instead requires information the radiograph may not contain. Adjacent vertebrae resemble one another, and the cue that separates them is position in a chain whose end is known; a two-body frame with no sacrum, no ribs and no cervical spine in it does not carry that cue. Whether such a frame carries enough residual morphology to be named, and how much of it survives on a spine with an extra or a missing segment, is an empirical question with a number attached.

Where that number is low the shortfall belongs to the exposure rather than to the classifier, and the remedy is the one a radiographer already performs: do not take one picture, take a series. A sweep of overlapping exposures from an unambiguous landmark builds a map of this patient's own spine, and a later limited field is matched onto it. That is an intraoperative registration built from radiographs rather than from a preoperative CT. This study measures what a single limited field can be named at, what the sweep adds, what it costs in exposures, and where it fails.

Materials and Methods

This retrospective study used public, de-identified data only; institutional review determined it not to be human-subjects research. Reporting follows CLAIM 2024 [22]. Every number below is produced by one script from named inputs with a hashed ledger.

Ground truth for border anomalies

Two independent ends of the column were labelled separately, because the study's hypotheses concern whether they travel together. For CT, 1,113 spines—802 from CTSpinoPelvic1K [23] and 311 from VerSe [24]—carried a lumbosacral flag (a sixth lumbar body by voxel count, or an adjudicated Castellvi grade [25]) and a thoracolumbar flag (a thirteenth thoracic body, a lumbar rib, or a rib-bearing count other than 12 where the field reached the twelfth rib), read from the label volumes rather than from any manifest, after a manifest T13 field was found to say 18 where the voxels said 6. For real radiographs, BUU-LSPINE [26] ships its authors' own per-study reading of four-lumbar (n = 129) or six-lumbar (n = 15) anatomy on 3,600 lateral and 3,600 anteroposterior films, together with four corner keypoints per lumbar body, two S1 end-plate corners on lateral films, and a pixel spacing of 0.175 mm. This sheet is the only radiograph-domain anomaly ground truth in the study and its negatives are contaminated by construction: a Castellvi I–IV segment on a five-lumbar spine is not recorded, so every specificity or AUC measured against it is a floor. Every anomalous spine from every source, including 211 decoder-flagged and 183 literature-figure candidates that were *not* used as truth, is listed with image dimensions, pixel size where stated, and four crop-level quality statistics in the released manifest (Fig. S1).

Imaging data and crop banks

Vertebra crops were formed at a fixed scale of 96 pixels per vertebral body on a 320×320 canvas with a target-marker channel, so that the pixel footprint of a body is constant across sources and the classifier cannot read size as identity [27]. Bank v2 held 105,960 crops from eight sources: three real-film collections with human or counted level names (CSXA [28], BUU-LSPINE, NHANES annotated subset [29]) and DRRs of CTSpinoPelvic1K and VerSe including physics-based renders of anomaly-enriched spines. Bank v5 added six sources for 253,763 crops, of which 76,042 (30.0%) were named by our own whole-spine template decoder on unannotated NHANES, VinDr-SpineXR [30], and lateral films and never verified by a reader. Table 1 states the label origin of every source (human 14.1%, carried from CT 45.8%, counting-inferred 10.2%, pseudolabel 30.0%) and the median sharpness, contrast, noise, and signal-to-noise ratio of its crops. Splits were patient-grouped; no film contributed to both partitions.

Model

The encoder was a ResNet-18 [31] with a 27-class output (C1–C7, T1–T12, T13, L1–L6, sacrum; T13 and L6 are variant classes that an ordinary spine skips), trained with cross-entropy, augmentation tier 2, batch 64, 40 epochs, seed 0. Alternative encoders and losses—CORN ordinal regression [32], ordinal focal loss, DINOv3 frozen and fine-tuned [33], a radiograph-pretrained DINO—were trained on the same recipe and scored on one common held-out set (Table 3). A run-level head, a bidirectional GRU over the frozen embeddings of 2–9 consecutive detected vertebrae, was trained on windows of mixed length [34]; it is the deployed system and produced every number in the Results. Each head decodes only the encoder it was trained on; the pairing is recorded by hash, because a mismatched pairing produced 0–1% accuracy with a systematic two-level offset that could pass for a finding.

Field-of-view states and the claims

Every held-out film was cut into eight windows of random length (2–9 vertebrae) at random offsets and each window was assigned, from its *true* labels, to the report's own states: A (cranial anchor: any cervical or rib-bearing thoracic level in frame), B (caudal anchor: sacrum in frame), or C (isolated stack: neither). Each was further split by whether the spine was anomalous under the ground truth above and by domain (real film or CT render). A film with no anomaly ground truth from either source was dropped from the

anomalous/ordinary split rather than assumed ordinary, because reading absence of evidence as absence of anomaly manufactures specificity; 2,258 of 2,927 v2 held-out films and 5,102 of 6,580 v5 films were so dropped. Those states are defined from labels, and that has a consequence stated here rather than discovered by a reader: BUU annotates L1–L5 with the sacrum present in the film, CSXA annotates C3–C7 with C2 present, and each crop is a band one body above and below its target, so a label-defined State-C window that touches either end of the annotated stack has the anchor in its pixels, and a window that is the whole stack has its names fixed by the annotation convention. A second evaluator (`tools/eval_interior_windows.py`) therefore scores only INTERIOR windows—those excluding the first and last annotated body—with a cluster bootstrap over films (2,000 resamples) and two structural baselines learnt from the training partition's label lists alone: a placement-blind prior that knows the collection and the window length, and a placement-aware prior that also knows the window's offset in the stack, which is what counting from the film's edge supplies. It also reports the share of held-out films whose patient stem appears in the training partition. Five claims were scored: C1, exact naming in State C by spine class; C2, per-level accuracy L1–L6 inside State C; C3, the signed ordinal error and the share of errored windows in which every error shared one offset; C4, the firing rate of the report's four continuous detectors on NHANES, VinDr, and AASCE [35] at thresholds fitted on CT renders; C5, State B as the ceiling.

The serial sweep, and the protocol behind it

A limited field cannot see a landmark; a series of limited fields can. The protocol is the one a radiographer already performs by hand, made explicit: start with the sacrum at the bottom of the field (or with C2 at the top), expose, move the C-arm one field minus one vertebral body, expose again, and repeat to the level of interest. The frames are linked into a single chain of vertebrae and stitched into one panorama, which then serves as the patient's own coordinate map—an intraoperative registration built from radiographs, with no preoperative CT anywhere in it.

Serial sweeps of one patient do not exist in any public collection, so a sweep was cut from each labelled long image: every CTSpinoPelvic1K render that reaches the sacrum (n = 801), the VerSe whole-spine renders swept from C2 downward, the real films that reach the sacrum, and 236 whole-spine lateral films. Each frame is a window of a stated physical span, taken with a jittered step, and each frame is degraded independently—gain, offset, gamma, a small rotation and scale, and the fitted film-noise model—so no two frames share pixels exactly. The frame span is physical, not a count of vertebrae: a 9-inch image intensifier is 23 cm across

with about 90% usable, a 12-inch one 31 cm, and a mobile C-arm's fixed 100 cm source-to-detector distance magnifies the spine by about 1.5, so the span at the spine is about 14, 19 and 20 cm for the 9-inch, 12-inch and flat-panel detectors and about 10 cm collimated [36]. At a lumbar pitch of 34 mm those hold about four, five and a half, six, and three vertebral bodies—the last is the report's own "2.5 to 3 segments".

The alignment is deliberately local. A radiograph integrates all depths, so under a translating cone-beam source no single two-dimensional mapping aligns every structure in two frames; parallax is a geometric constraint and not an engineering nuisance, and the published remedies—cancelling the translation with a tracked table, or reconstructing an orthographic mosaic—need hardware a freehand sweep does not have [37]. Consecutive frames are therefore linked by a translation estimated on the overlap band alone, never by a global homography, and the panorama is a display and a search space rather than a metric reconstruction. That translation is found by normalised cross-correlation restricted to within 0.6 of a body of the step the protocol asked for, because the column is periodic and an unrestricted correlation locks one body off—the single documented failure mode of automatic vertebral localisation, and the one that accounted for all 47 failures in the original 3-D/2-D registration study [38]; the shared bodies follow from the translation, and the pairing the encoder's embeddings alone would have chosen is recorded beside the one used. The chain is named twice: by counting from the declared landmark with the ordinary alphabet, and by the run-level head over the whole chain with the landmark declared and the variant columns open. A later limited-field frame is registered to the panorama and takes its names from the nodes it lands on. Chain integrity—duplicated, missed and spurious nodes—is scored separately from naming, so a naming figure that rests on a broken chain is visible as such. What the simulation cannot reproduce is parallax between exposures and patient motion between them; both make the linking harder and neither changes the naming, and the linking is reported on its own for that reason.

Statistical analysis

Accuracies carry Wilson 95% CIs on the vertebra; the unit of clustering (film) is acknowledged as a limitation and film counts are reported beside every stratum. Chance was computed from the scored label distribution ($\sum_c p_c^2 = 10.3\%$), not as $1/27$. No hypothesis test was applied to accuracy differences; non-overlapping intervals are described as such.

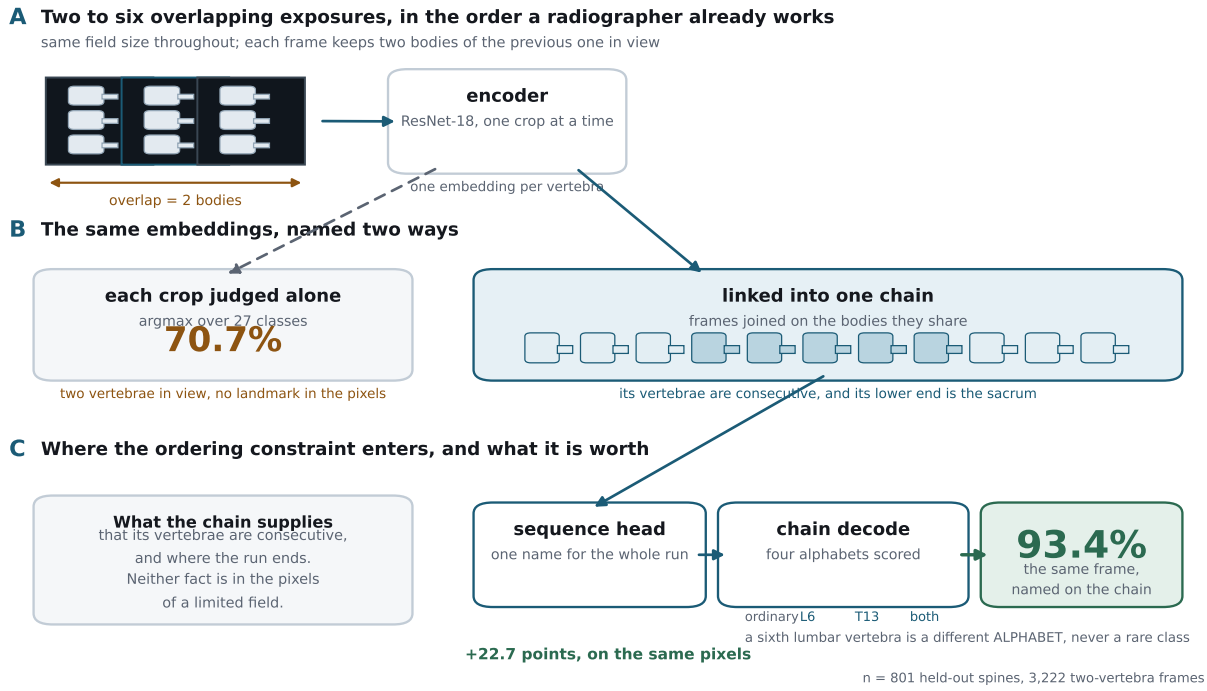


Figure 1: Overlapping exposures are linked on the bodies they share into one chain, and the same per-vertebra posteriors are then decoded under the constraint that the chain is consecutive and ends at a landmark. The accuracies are read from the result file, not typed.

Results

Every figure is measured on spines the system was never fitted on, with the exclusion enforced by the evaluation tool rather than by hand. Samples are small in consequence and the intervals are correspondingly wide.

One limited field, named twice

The central comparison is one set of pixels named two ways: on the frame's own posterior, and with the chain's ordering constraint applied (Fig. 1, Table 2). On 120 held-out abdominopelvic renders at a 9-inch field, a two-vertebra frame carrying no landmark was named exactly 72.1 % of the time alone and 95.2 % on the chain; at three vertebrae 86.8 % and 94.7 %; at four, 90.2 % and 94.8 %. On 25 held-out real lumbar radiographs the same comparison is 78.7 % and 93.6 % at two vertebrae, and 85.7 % and 93.9 % at three. Every spine in these arms is one the encoder never saw: 681 of the 801 renders, 145 of the 170 films and 64 of the 76 whole-spine studies lie in the encoder's training split and are refused by the tool, and the numbers first reported over all of them were withdrawn for that reason. A second set of render numbers was withdrawn

after it: the renderer had written the anterior–posterior voxel size as the row spacing, so the "140 mm" field was 7.5 % too large at the median on the abdominopelvic renders and, on the whole-spine renders with 1.25 mm slices, four times too large—two frames the height of the spine. Every render figure here is on frames of the stated physical size; the real films carry no header spacing and were unaffected.

The effect is largest exactly where a single frame has least to work with. Swept downward from C2 on whole-spine studies, a two-vertebra thoracic frame was named 9.5 % alone and 100 % on the chain, a three-vertebra frame 25.0 % and 100 %, from 2.9 exposures with every chain complete. Those vertebrae genuinely resemble one another, which is why counting from a landmark is the standard practice there and why counting errors grow with distance from it; a match onto a patient-specific map does not care how far the frame sits from the anchor. That arm rests on 12 held-out spines and the direction, not the value, is what it establishes.

Counting, made exact

A surgeon does not classify a vertebra from a limited field. They count: down from C2, or up from the sacrum, to the level on the consent form. The failure is not that counting is the wrong idea but that it is performed across several exposures, each of which can be joined to the last one body off. That suggests a question separate from everything above, and with a different kind of answer: if the chain carries the count rather than a name, how exact is it?

The arm that asks this uses no learned component. Frames are cut by the same protocol and linked by the same image correlation, but no network is loaded; every node is named by its ordinal from the anchor, and the target is the k-th body from it. Success is therefore ordinal—the mark is on the body whose position in the labels is k—and variant anatomy never enters, because no alphabet is chosen. Because nothing is fitted, no spine has to be withheld, and every scan carrying the anchor is used.

Counting down from C2 was exact. On 76 whole-spine studies, walking from C2 to every labelled body in turn, 832 of 834 walks put the mark on the right vertebra at two bodies of overlap and **834 of 834 at three**, from 2.2 exposures; the two failures at two bodies were one cervical study whose 1.25 mm slices give a 12-pixel vertebral pitch, where a link paired nothing. On 204 TotalSegmentator studies that hold C2—a corpus this work has never trained on in any form—2,646 of 2,654 walks that reached their target were exact (99.7 %), the misses confined to two spines. Counting up from the sacrum was almost as good and failed differently: 99.66 % of the 6,255 reached walks on all 801 abdominopelvic spines and 99.82 % of 7,181 on

522 TotalSegmentator spines, with the errors concentrated in six and three spines respectively, one of which carries a label error in the release itself. What does not reach is not a failure of counting but of field: 97 walks on the abdominopelvic renders and 215 on TotalSegmentator ended before the target appeared, almost all of them the topmost labelled body, or a cervical level 18 to 19 bodies from the sacrum, where the image runs out first.

Two things follow, and the second matters more than the first. Overlapping fields turn counting from an approximation into an exact operation, at two to three exposures, provided the anchor is in the first field and the vertebrae are found in each one. And the second condition is where the residual risk now sits: these numbers use the annotation's boxes, so the only thing that can fail is the link, and it almost never does. A missed or spurious box, by contrast, shifts the ordinal by one and renames every body above it—which is the same failure the surgeon makes, reintroduced by a detector. The naming network is what remains when counting cannot be trusted: when the anchor is out of the field, when the image runs out first, and when the spine has an extra or a missing segment, so that the count reaches a body the ordinary alphabet has no name for.

What the chain costs, and where it does not help

The chain came out complete on 97.5 % of held-out CT spines and on every held-out real radiograph, at two bodies of overlap and 3.4 exposures. Overlap has a knee: half a body completes 81.7 % of chains at 2.9 exposures, one body 95.8 % at 2.9, two bodies 97.5 % at 3.4, three bodies 90.8 % at 4.8, and four bodies falls back to 82.5 % while costing 9.2, because every extra frame is another link that can be joined wrongly. Field size moves it less than overlap does: at one body, a collimated field completes 92.4 %, the 9-inch field 95.8 % and a 12-inch field 99.2 % from 2.0 exposures. Two bodies on the 9-inch field is the operating point and it survived re-measurement at the corrected field size.

The protocol as written reproduces it, once its last move is stated. Every arm above simulated the sweep with a step jittered by ± 0.3 body. The instruction a radiographer is handed is a fixed move—the sacrum at the bottom of the first field, then one field minus two bodies each time—and that fixed step was run on the same 120 spines with the same photometric draws, so only the geometry differed. It completed 95.0 % of chains against 97.5 %, and every spine it lost failed on the last frame: the simulation slid the final field to fit the top of the scan, so its move was whatever was left, 0 to 6 pixels on those six spines—a frame no radiographer would take and one the linker, which refuses steps under half a body to keep the periodic column from aliasing,

cannot represent. Two remedies were run. Refusing a final move shorter than half a body completed 120 of 120 chains, with a limited field named on the chain at 94.7 %, 96.3 % and 94.2 % at two, three and four vertebrae; letting the last link alone search down to a zero step, since it is the only link the protocol allows to be short, also completed 120 of 120—and did the same for the jittered reference, whose three incomplete chains were the same final frame. With the end of the sweep handled, no chain on the 120 held-out renders is incomplete at two bodies of overlap under either step. The protocol therefore carries one more line than it did: the last move is a full move or none. The number a surgeon would get is the number reported (Table 5).

Three changes from panoramic stitching were tried against it and none moved the operating point. Panorama software does not chain frames pairwise; it solves every frame's position at once so that a wrong link is outvoted by its neighbours, seeds each match with phase correlation, and refines the placement by direct alignment. Each was added to the tool and run on the same 120 spines, alone and together (Table 5). The global solve does what it is for where it has something to work with: at three and four bodies of overlap, where a frame shares content with the frame after next, chains complete on 91.7 % and 90.0 % of spines against 90.8 % and 83.3 %, and duplicated vertebrae at four bodies fall from 0.12 to 0.07 per spine. At two bodies of overlap on a four-body field, frames two apart share nothing, and the solve cost three spines (92.5 % against 95.0 %). The phase-correlation seed changed no chain; the direct-alignment refinement changed no number to the third decimal, because the correlation peak it starts from is already at the right shift. Naming on the chain moved by less than a point in every arm but the global solve at two bodies, where it fell two to three. The fixed two-body step remains the protocol, and the global solve is kept in the tool for the higher-overlap sweeps a radiographer might take by choice.

A second bench, ten times larger, that no model here has seen. TotalSegmentator's public release is 1228 routine clinical CTs from one institution's practice—abdomen–pelvis, thorax–abdomen–pelvis, polytrauma, neck—with a vertebra label per level, and not one of them is in any training bank of this work. Rendered the same way, 1170 carry a usable label stack (the rest have gaps, the label mix-ups the release notes describe); 522 reach the sacrum and 203 hold C2. Its labels carry no L6 or T13 class, so it is an ordinary-spine bench and says nothing about variant naming. Swept from the sacrum with the protocol as written, chains completed on 98.5 % of the 522 from 4.2 exposures, and a limited field was named on the chain at 94.3 %, 95.0 % and 94.4 % at two, three and four vertebrae, from 51.2 %, 72.7 % and 82.7 % alone—a larger single-frame deficit than the abdominopelvic renders show, because these frames are mostly thoracic. Swept from C2 on the 203 scans that hold it (8 to 24 bodies each), chains completed on 95.1 % from 2.8 exposures, and a two-vertebra

frame went from 18.9 % to 94.0 %, a three-vertebra frame from 39.5 % to 96.4 %, and a four-vertebra frame from 58.8 % to 97.8 % (Table 2). Before the end-of-sweep rule the sacrum-anchored figure was 91.4 %, and every one of the 48 lost chains had a final clamped step of one to ten pixels, the same artefact.

A perfect rib reader would not have helped. Whether a body bears a rib is what places the thoracolumbar junction on a film, and the one real-film decoder error above is exactly a junction placed one body too low, so the ceiling of that cue was measured before building anything: rib-bearing truth from the CT labels—CTSpinoPelvic1K's per-level ribs and TotalSegmentator's 24 named ribs—entered the alphabet decoder as a soft penalty per body whose name disagrees with its rib. No arm improved. At two bodies of overlap the term cost one of 120 spines, whose twelfth rib is absent, by making the L6 alphabet the rib-consistent story; at four bodies it changed one false T13 call into a false L6; on the 522 TotalSegmentator spines it moved naming on the chain by 0.3 points. The shape reader and the count already resolve the junction on a CT chain, and rib count is itself variant anatomy—an absent or lumbar twelfth rib is common enough that a rib reader is another uncertain reading, not an arbiter. The film error stays open, and it is stated as one film in 25.

With the detector's own boxes, the real films. Everything above uses the annotation's boxes, so it measures the linker and the decoder. Run with the film detector's boxes on the 25 held-out films, the chain first completed on 10 of 25. Rendered, 12 of the 15 breaks were the same thing: the detector had found the T12 or T11 that sits above L1 in the film, the system had named it T12 or T11, and the scorer had counted a body the label set does not cover as a false detection. A box with no truth at the far end of the chain from the anchor is now recorded as unlabelled, with the name it was given, and not held against the chain; 13 such bodies were named T11 or T12 across the 25 films. So scored, the detector's chains complete on 21 of 25 films, the anchor is found on 23, and a limited field is named on the chain at 87.2 %, 90.4 % and 91.4 % at two, three and four vertebrae against 93.6 %, 93.9 % and 100 % with true boxes. Two linker changes built for detector boxes—ordering chain nodes by their panorama position and merging boxes from different frames that coincide—changed no number on the films, and merged distinct bodies on true-box renders, so they are off. Of the four films still broken, two are a single missed body, one is a missed sacrum, and one is the decoder: on a film showing eight bodies above the sacrum the count read T11 to L5 correctly, but the shape reader placed every body one level caudal with high confidence and the L6 alphabet won by a margin (0.74) inside the range of the genuine T13 calls on renders (0.46–0.85), so no threshold separates them and scoring only the disputed bodies does not help, because all seven are disputed. Only the count or a rib settles that film, which is the case for a rib-bearing cue and is left open here.

On renders the detectors fail before the linker. A detector trained on the renders and holding a sacrum class found the anchor on six of eight spines held out from both it and the encoder, where the radiograph-trained detector found it on four; neither held a chain together on any of the eight, because their boxes on adjacent frames did not correspond. Eight spines is what survives two holdouts, and it is reported as eight. The earlier detector-box figures on renders were also wrong for a second reason: the tool read only pose keypoints, so a detection model returned no boxes at all, and "chain complete 0 %" measured the tool.

One qualification belongs beside the real-film numbers and is the reason the claim is narrow. Those 25 films reach the sacrum, and counting up from the sacrum alone names every one of those chains. With the landmark in view the sweep matches a count rather than beating one. What it adds is the naming of frames that cannot see the landmark, which is the 78.7-to-93.6 and 85.7-to-93.9 columns, and that is the whole of the claim.

Per level, per spine class, and where it fails

Table 2 gives the per-class breakdown. It is not 99 % everywhere. On the chain with three vertebrae in the frame, ordinary spines scored 90.5–92.7 % at L1 to L4, 95.1–96.4 % at T11 and T12 and 100 % at L5, the level beside the anchor; the six lumbosacral-variant spines that survive the holdout were named correctly at every level, and six is too few to say more than that. The held-out set holds a single sixth lumbar vertebra and it was named L5 alone and on the chain; what is known about L6 is the encoder's 53–74 % in Table 3, and the sweep has not been shown to rescue it. Ninety-nine percent is reachable only by declining, and only at three vertebrae: ranking three-vertebra frames by the decode's own probability, the top 80 % were named correctly 99.5 % of the time; four-vertebra frames reach 98.3 % on their top 80 % and no higher at any coverage, and ranking by the probability times the weakest link's margin reaches 98.2 % on the top half.

Two failures are structural rather than statistical. Every transitional cell rests on tens of vertebrae, not thousands. And a thirteenth thoracic vertebra was named correctly by no encoder arm on the common held-out set (Table 3), because the corpus contains almost no examples of one—a data limitation stated as such.

Which encoder, on one common set

Fourteen encoder arms were scored on the 2,927 patients in the validation split of every bank and the training split of none (Table 3). The loss matters more than the corpus: the three logit-adjustment arms lead at 80.8–81.1 % exact and lie within 0.3 points of one another across five bank versions, while every cross-entropy and ordinal arm falls between 77.1 and 79.9 %. The single exception is the class the work exists for—the arm trained on the bank that first contained real photographs of six-lumbar spines is the worst overall (77.1 %) and the best at L6 (73.5 % against 53–67 %). That trade is the result, not a tuning artefact.

Against what is published

No two systems in this field report the same quantity, so Table 4 gives each published figure beside a statement of what was measured. Registration convergence, geometric label placement conditioned on a correctly annotated CT, a surgeon questionnaire, detection F_1, Panoptic Quality and a CT identification rate are six different measurements, and only one published number is a registration-free, radiograph-only, per-vertebra naming accuracy: 88.3 % on 13 cadaveric specimens at a full 64 cm field, from the laboratory that built the registration system it is usually contrasted with [39]. The widely quoted 97.6 % is five surgeons reading with an overlay, not a machine figure [40].

Discussion

A limited fluoroscopic field with no landmark in it does not contain the information needed to name the vertebra it shows. That is not a statement about model capacity; it is a statement about the image. Adjacent vertebrae resemble one another, and the cue that separates them—position in a chain whose end is known—is not in the pixels of a two-body frame. A short sweep supplies exactly that missing cue and nothing else: the frames are linked on the bodies they share, which establishes that the vertebrae are consecutive and where the run terminates, and the same posteriors are then decoded under that constraint. The gain, 23 points at two vertebrae on renders and 84 points in the thoracic spine, is the size of the information that was missing.

Two properties of the failures matter more than the headline. When the system erred without an anchor it did so by a one-level shift 83 % of the time, and in 45 % of errored windows as a single coherent offset across the whole window—which is the failure mode a chain decode can detect and a per-frame classifier cannot, because a coherent offset is exactly what a chain scores badly. And the variant class is where the method is

weakest: a sixth lumbar vertebra is named correctly less than four times in five even on the chain. A surgeon counting from a landmark obtains neither number, because counting has no posterior.

Why this cannot be ranked against registration. A registration algorithm never names a vertebra. It solves a six-degree-of-freedom pose, and the name is transferred from a human's reading of the preoperative volume—which is why the clinical report of 100 % geometric accuracy is stated as conditional on a correctly annotated CT [20, 21]. Two failures therefore survive a perfect registration: a miscounted preoperative scan propagates onto the radiograph faithfully and confidently, which is the failure mode of a transitional spine and the population in which wrong-level surgery occurs; and the surgeon must still accept or reject the overlay. The honest comparison is not machine against machine but machine against the count a surgeon would otherwise make, and for the assisted figure a reader study, which is running and is not reported here.

What remains. The corpus is the binding constraint and it is a constraint on people, not on parameters. Counted in distinct patients, the training partition holds thousands at each ordinary lumbar level and fewer than two hundred with six lumbar vertebrae, of which—until the most recent bank—every one was a rendering of a CT rather than a photograph. Every long-tail remedy applied to that imbalance measured as doing nothing; the one intervention that moved the class was the arrival of real photographs of it. Two studies follow from that and neither is done: a retrospective archive collection enriched for transitional anatomy and read against cross-sectional imaging, which would supply the first real transitional films in the intended domain, and only then a prospective study in theatre measuring naming, exposures and reading time against the surgeon's own determination.

Limitations. Anomaly ground truth on real film came from one collection and its authors' reading without cross-sectional confirmation; its negatives contain Castellvi I–IV segments, so anomalous accuracy is measured against a contaminated ordinary class and every geometric AUC is a floor. The anomalous real-film test set was 20 films in v2 and 25 in v5, of which only 12 and 17 contribute interior windows, so the anomalous interior intervals span 29–87%; the two encoder banks share those films' partition. The label-defined strata (Fig. 2, Table S1) pool whole-stack and end-touching windows and overstate anchor-free naming; the interior figures supersede them. Confidence intervals are on the vertebra and ignore clustering by film. VerSe is enriched for pathology and its base rates are not prevalences. The separability and signature probes used renders, not radiographs. The four continuous detectors were scored at CT-fitted thresholds only; recalibration on film was not attempted. Results dependent on VinDr-SpineXR and AASCE require the reader's own data-use agreement to reproduce. The system's L6 accuracy of 45–48% is a limitation of

the product, not of the study, and is reported as such. The sweep is simulated from single long images: it reproduces independent exposure, noise, rotation and scale between frames but not parallax between frames taken at different gantry positions, nor patient motion, and both act on the linking rather than on the naming. Every lateral film in BUU-LSPINE is a left lateral, right-lateral studies having been mirrored by its authors, so any handedness-dependent geometry in the two-view fusion is ambiguous by construction.

Tables

Table 1: The training corpus, counted in distinct patients rather than crops, because a rare class is learned from people and one spine is rendered from many poses. *Pixels* distinguishes a photograph from a rendering of a CT; *name from* distinguishes a name taken from a labelled volume, from a collection's own annotation, and from a decoder's output. Generated by `tools/fig_dataset.py`.

Source	Pixels	Name from	Patients	Crops	Patients by region
VinDr-SpineXR, pseudolabelled	film	decoder	6,475	36,622	lumbar 4,552, thoracic 2,142, cervical 1,908, sacral 945
CT renders, physics model	render	volume	5,918	31,628	lumbar 4,734, thoracic 2,971, cervical 678, variant 121
cervical radiographs	film	dataset	4,219	21,095	cervical 4,219
BUU-LSPINE lumbar radiographs	film	dataset	3,063	15,310	lumbar 3,063
fluoroscopy panels, pseudolabelled	film	decoder	2,028	12,648	lumbar 1,502, thoracic 847, cervical 533, sacral 386, variant 145
CT renders, variant-enriched	render	volume	1,217	6,258	lumbar 1,104, thoracic 539, variant 512, cervical 4
CTSpinoPelvic1K renders	render	volume	681	6,091	lumbar 681, sacral 681, thoracic 679, variant 17
NHANES renders	render	volume	464	2,608	cervical 234, lumbar 230, thoracic 184
NHANES II radiographs	film	dataset	463	2,741	cervical 237, lumbar 226, thoracic 187, sacral 145
VerSe renders	render	volume	318	3,837	thoracic 292, lumbar 259, cervical 113, variant 50
total			**24,671**	**138,838**	

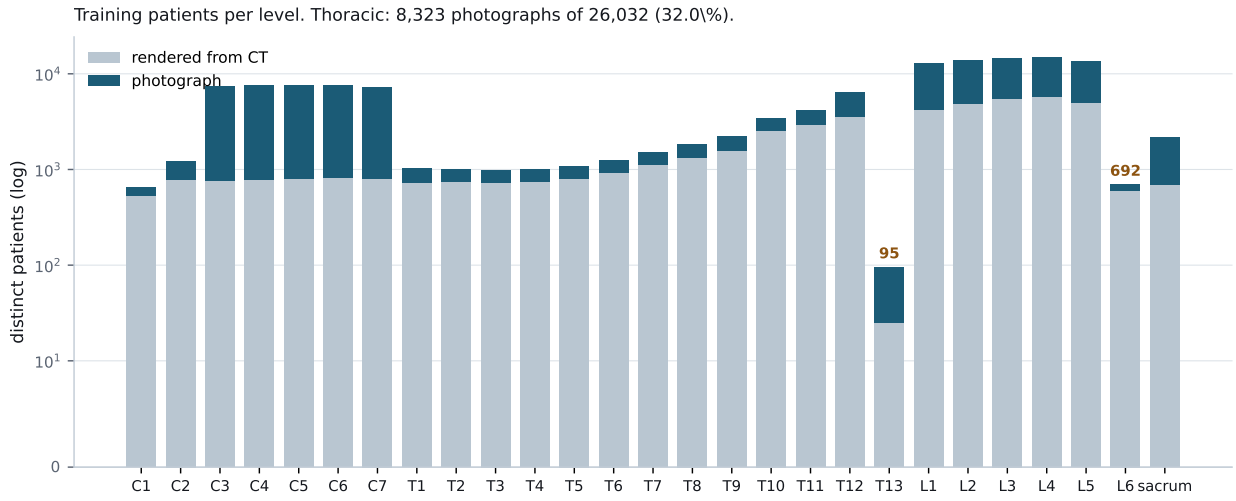


Figure 2: Training patients at each level, split by whether the image is a photograph or a rendering of a CT. The vertical scale is logarithmic. The shape of this figure, not the size of the corpus, is what every limitation in this paper is downstream of.

Table 2: One limited field, named twice: on the frame's own posterior, and with the chain's ordering constraint applied. The pixels are identical in both columns of each pair; only the constraint differs. n is held-out spines. Generated by `tools/make_results_table.py`; nothing in it is typed.

			2 vertebrae		3 vertebrae		4 vertebrae	
Arm	Spine	n	alone	chain	alone	chain	alone	chain
CT renders, 9-inch field	all	120	72.1	**95.2**	86.8	**94.7**	90.2	**94.8**
CT renders, 9-inch field	ordinary	75	69.9	**94.4**	85.5	**92.8**	89.2	**94.5**
CT renders, 9-inch field	lumbosacral	6	76.9	**100.0**	72.7	**100.0**	90.9	**100.0**
CT renders, 9-inch field	both	1	100.0	**100.0**	100.0	**100.0**	100.0	**100.0**
CT renders, 9-inch field	unknown	37	74.3	**95.9**	90.8	**97.4**	91.4	**94.3**
TotalSegmentator renders	all	522	53.3	**91.7**	71.4	**91.5**	82.8	**92.5**
TotalSegmentator renders	unknown	522	53.3	**91.7**	71.4	**91.5**	82.8	**92.5**
real radiographs	all	25	78.7	**93.6**	85.7	**93.9**	91.9	**100.0**
real radiographs	unknown	25	78.7	**93.6**	85.7	**93.9**	91.9	**100.0**
whole spine, swept from C2	all	12	9.5	**100.0**	25.0	**100.0**	30.4	**100.0**
whole spine, swept from C2	ordinary	3	25.0	**100.0**	33.3	**100.0**	40.0	**100.0**
whole spine, swept from C2	unknown	8	6.2	**100.0**	25.0	**100.0**	25.0	**100.0**
TotalSegmentator, swept from C2	all	203	18.9	**94.0**	39.5	**96.4**	58.8	**97.8**
TotalSegmentator, swept from C2	unknown	203	18.9	**94.0**	39.5	**96.4**	58.8	**97.8**

Figure Legends

Figure 1. The report's argument, drawn. The same field of view at the same height above the sacrum holds L4–L2 in an ordinary spine and L5–L3 in a spine with a lumbarised S1; the report concludes that absolute naming is underdetermined. The figure is the claim, not data.

Figure 2. Exact vertebra naming by the report's field-of-view state, split by real film versus CT render and by anomalous versus ordinary spine, with Wilson 95% CIs and vertebra counts. Left, bank v2; right, bank v5 (30% pseudolabelled crops). Dotted line, chance for the scored label distribution.

Figure 3. Per-level accuracy inside the isolated stack, L1–L6, both banks. Shaded, the levels the report called exchangeable.

Figure 4. Left, signed ordinal error of every isolated-stack error (v2); ± 0.5 is a variant class confused with its neighbour. Right, the shape of error within a window.

Figure 5. Balanced accuracy of a linear probe for four spine classes against window length, for windows containing neither junction and windows containing one, each with its shuffled-label control. Marker size is proportional to window count.

Figure 6. Left, AUC of each geometric index on real lateral and anteroposterior films for six-lumbar and four-lumbar spines. Right, the Wigh index of the lowest body on 3,600 lateral films with the report's "normal

Table 3: Fourteen encoder arms on one common held-out set: the 2,927 patients in the validation split of every bank and the training split of none, so the banks are comparable. Exact and macro accuracy over 27 classes, with the four levels that separate the arms. Generated; nothing is typed.

Arm	exact	macro	T12	L5	L6	T13
r18_la_v2	81.1	71.3	75.1	83.4	53.0	0.0
r18_la_v6	81.0	70.8	75.0	84.2	62.9	0.0
r18_la_v7	80.8	71.0	73.9	79.6	72.0	0.0
r18_ce_v4	79.9	70.7	75.8	83.1	55.3	0.0
r18_ce_v3	79.7	71.0	75.7	82.2	59.1	7.1
r18_ce_v5c	79.1	70.2	74.9	80.2	55.3	0.0
r18_ce_v6	79.0	70.8	76.8	80.6	67.4	0.0
r18_corn_v6	78.9	70.5	75.8	80.4	62.1	0.0
r18_ce_v5	78.8	70.7	76.1	76.3	62.9	0.0
r18_ce	78.8	70.4	71.2	78.9	64.4	0.0
r18_corn_v5	78.7	70.6	75.4	76.8	62.1	0.0
r18_corn_v5c	78.7	69.9	74.7	77.8	57.6	0.0
r18_corn	78.4	70.0	76.9	79.9	60.6	0.0
r18_ce_v7	77.1	69.7	71.1	74.2	73.5	0.0

Table 4: Published figures beside a statement of what each one measures. This is deliberately not a ranking: registration convergence, label placement conditioned on a correct CT, a questionnaire, a detection score and a naming accuracy are different quantities, and the third column is the point of the table.

Method	Setting	What was measured	Value	n	95% CI
Naming one limited field, with and without the chain		The same pixels, named twice: on the frame's own posterior, then with the chain's ordering constraint applied. This is the paper's central comparison.			
this work, CT renders	2 vertebrae, all	exact name, frame alone	72.1	484	
this work, CT renders	2 vertebrae, all	exact name, on the chain	95.2	484	[91.6, 98.4]
this work, CT renders	3 vertebrae, all	exact name, frame alone	86.8	720	
this work, CT renders	3 vertebrae, all	exact name, on the chain	94.7	720	[90.8, 98.2]
this work, CT renders	4 vertebrae, all	exact name, frame alone	90.2	904	
this work, CT renders	4 vertebrae, all	exact name, on the chain	94.8	904	[90.8, 98.2]
this work, CT renders	the whole chain, all	exact name	96.7	120	[93.8, 99.1]
this work, CT renders	2 vertebrae, ordinary	exact name, frame alone	69.9	302	
this work, CT renders	2 vertebrae, ordinary	exact name, on the chain	94.4	302	[89.0, 98.4]
this work, CT renders	3 vertebrae, ordinary	exact name, frame alone	85.5	447	
this work, CT renders	3 vertebrae, ordinary	exact name, on the chain	92.8	447	[86.9, 97.4]
this work, CT renders	4 vertebrae, ordinary	exact name, frame alone	89.2	564	
this work, CT renders	4 vertebrae, ordinary	exact name, on the chain	94.5	564	[88.9, 98.6]
this work, CT renders	the whole chain, ordinary	exact name	95.9	75	[91.9, 99.0]
this work, CT renders	2 vertebrae, lumbosacral	exact name, frame alone	76.9	26	
this work, CT renders	2 vertebrae, lumbosacral	exact name, on the chain	100.0	26	[100.0, 100.0]
this work, CT renders	3 vertebrae, lumbosacral	exact name, frame alone	72.7	33	
this work, CT renders	3 vertebrae, lumbosacral	exact name, on the chain	100.0	33	[100.0, 100.0]
this work, CT renders	4 vertebrae, lumbosacral	exact name, frame alone	90.9	44	
this work, CT renders	4 vertebrae, lumbosacral	exact name, on the chain	100.0	44	[100.0, 100.0]
this work, CT renders	the whole chain, lumbosacral	exact name	100.0	6	[100.0, 100.0]
this work, CT renders	2 vertebrae, thoracolumbar	exact name, frame alone	100.0	4	
this work, CT renders	2 vertebrae, thoracolumbar	exact name, on the chain	100.0	4	[100.0, 100.0]
this work, CT renders	3 vertebrae, thoracolumbar	exact name, frame alone	100.0	6	
this work, CT renders	3 vertebrae, thoracolumbar	exact name, on the chain	100.0	6	[100.0, 100.0]
this work, CT renders	4 vertebrae, thoracolumbar	exact name, frame alone	100.0	8	
this work, CT renders	4 vertebrae, thoracolumbar	exact name, on the chain	100.0	8	[100.0, 100.0]
this work, CT renders	the whole chain, thoracolumbar	exact name	100.0	1	[100.0, 100.0]
this work, CT renders	2 vertebrae, both	exact name, frame alone	100.0	4	
this work, CT renders	2 vertebrae, both	exact name, on the chain	100.0	4	[100.0, 100.0]
this work, CT renders	3 vertebrae, both	exact name, frame alone	100.0	6	
this work, CT renders	3 vertebrae, both	exact name, on the chain	100.0	6	[100.0, 100.0]
this work, CT renders	4 vertebrae, both	exact name, frame alone	100.0	8	
this work, CT renders	4 vertebrae, both	exact name, on the chain	100.0	8	[100.0, 100.0]
this work, CT renders	the whole chain, both	exact name	100.0	1	[100.0, 100.0]
this work, CT renders, 1-body overlap	2 vertebrae, all	exact name, frame alone	72.7	476	
this work, CT renders, 1-body overlap	2 vertebrae, all	exact name, on the chain	95.2	476	[91.6, 98.1]
this work, CT renders, 1-body overlap	3 vertebrae, all	exact name, frame alone	84.7	720	
this work, CT renders, 1-body overlap	3 vertebrae, all	exact name, on the chain	93.6	720	[89.3, 97.5]
this work, CT renders, 1-body overlap	4 vertebrae, all	exact name, frame alone	90.1	912	
this work, CT renders, 1-body overlap	4 vertebrae, all	exact name, on the chain	95.7	912	[92.1, 98.7]
this work, CT renders, 1-body overlap	the whole chain, all	exact name	96.7	120	[94.0, 99.0]
this work, CT renders, 1-body overlap	2 vertebrae, ordinary	exact name, frame alone	72.1	294	
this work, CT renders, 1-body overlap	2 vertebrae, ordinary	exact name, on the chain	94.2	294	[89.0, 98.0]
this work, CT renders, 1-body overlap	3 vertebrae, ordinary	exact name, frame alone	85.2	459	
this work, CT renders, 1-body overlap	3 vertebrae, ordinary	exact name, on the chain	92.6	459	[86.4, 97.6]
this work, CT renders, 1-body overlap	4 vertebrae, ordinary	exact name, frame alone	89.6	556	
this work, CT renders, 1-body overlap	4 vertebrae, ordinary	exact name, on the chain	96.0	556	[91.7, 99.3]
this work, CT renders, 1-body overlap	the whole chain, ordinary	exact name	96.0	75	[92.3, 98.9]
this work, CT renders, 1-body overlap	2 vertebrae, lumbosacral	exact name, frame alone	58.3	24	
this work, CT renders, 1-body overlap	2 vertebrae, lumbosacral	exact name, on the chain	91.7	24	[75.0, 100.0]
this work, CT renders, 1-	3 vertebrae, lum-	exact name, frame alone	75.0	36	

Table 5: The protocol as written and what was tried on it. Every row is the same 120 held-out CT renders with the same photometric draws; the first row is the jittered-step reference, the second the fixed sacrum-to-cranial step a radiographer is told to make, and the rest add one panoramic-stitching change at a time at the stated overlap. Generated; nothing is typed.

Arm	overlap	exposures	chain complete	duplicates/spine	2 vert.	3 vert.	4 vert.
jittered step (reference)	2	3.42	97.5	0.000	95.2	94.7	94.8
fixed step, as written	2	3.39	95.0	0.042	93.6	94.7	93.9
fixed step, final move under half a body refused	2	3.32	100.0	0.000	94.7	96.3	94.2
fixed step, last link searched to zero step	2	3.39	100.0	0.000	94.4	95.4	94.9
jittered step, last link searched to zero step	2	3.42	100.0	0.000	95.2	95.1	95.0
+ global link solve	2	3.39	92.5	0.058	92.1	92.5	92.1
+ phase-correlation seed	2	3.39	95.0	0.000	94.0	94.3	94.1
+ ECC placement	2	3.39	95.0	0.042	93.6	94.7	93.9
+ all three	2	3.39	92.5	0.067	91.3	91.7	91.6
fixed step	3	4.80	90.8	0.033	95.0	92.8	95.0
+ global link solve	3	4.80	91.7	0.083	93.2	91.4	92.4
fixed step	4	9.43	83.3	0.117	93.9	94.6	95.2
+ global link solve	4	9.43	90.0	0.067	94.1	94.4	94.4
+ all three	4	9.43	90.8	0.050	94.4	94.4	94.7

L5” band shaded.

Figure 7. The serial sweep. Left, exact naming against the number of vertebrae in a limited-field frame, for each spine class: dashed, the frame named on its own pixels; solid, the same frame registered onto the sweep. Right, the same comparison against the physical field of each swept frame, with the share of spines whose chain of vertebrae came out complete printed under each. Drawn by `tools/make_sweep_figure.py` from `SWEEP_TABLE.json`.

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